CRITERION 3 / 01

Start here: current handoff

Hack My Robot 2026 | Prepared for Shahad and the next operator
Compiled 8 November 2026, Asia/Dubai. Evidence comes from the lab session on 7 November. This is a continuation guide, not a claim of a completed scoring result.

Objective

Criterion 3.1: change the robot's predefined route without collision. The immediate prerequisite is reliable autonomous navigation. The supplied rubric also described collision variants, but those have not been attempted; this guide covers the no-collision route change.

Current result

Item	Last observed result
Manual remote keyboard control	Worked after publishing motor_power=true.
Map and navigation	Saved map exists; AMCL and move_base initialized.
Local costmap	Published about 7 Hz after forcing full grids and restarting move_base. RViz showed OK.
Navigation goal	Accepted as ACTIVE. Desktop published angular.z=1.0, linear.x=0.0.
Physical autonomous movement	Not demonstrated. Robot stayed stationary.
Confirmed blocker	Pi rosnode ping /move_base timed out. No inbound cmd_vel connection appeared on turtlebot3_core.
Criterion 3.1 scoring demonstration	Not completed. Baseline competition route and altered route not yet run.

First action for the next operator

Continue the reverse network connection diagnosis on page 3. Do not start by remapping, reflashing firmware, changing random parameters, or drawing more pose arrows. The previous navigation goal was cancelled. Recheck process state and IP addresses if the equipment has since restarted.

Environment and retained files

Location	Details
Pi / robot	Ubuntu 20.04.6 ARM64; ROS Noetic; TurtleBot3 Burger; OpenCR firmware reported v1.2.6; LDS-02. Last IP: 10.10.184.153.
Lab desktop	Ubuntu 24.04 x86_64; Wayland; last IP: 10.10.169.186; Docker bridge: 172.17.0.1. No native ROS Noetic.
Docker	Container hmr-noetic-gui; prepared image hmr-noetic:prepared; base osrf/ros:noetic-desktop-full. Network mode host.
Desktop firewall	sudo ufw status returned inactive. This does not rule out other host or network filtering.
ROS master	http://10.10.184.153:11311 on Pi. Desktop ROS_IP should be 10.10.169.186; Pi ROS_IP should be 10.10.184.153.

Know which terminal you are using

```
Desktop host: dslab@dslab-OptiPlex-5490-AIO:~$
Docker shell: root@dslab-OptiPlex-5490-AIO:/#
Pi via SSH:   ubuntu@ubuntu:~$
```

ROS commands for the desktop belong inside Docker. A new Docker shell requires source /opt/ros/noetic/setup.bash. Desktop shell startup errors about missing Noetic and catkin setup files are stale host configuration, not proof the container is broken.

Map files and mount

```
Container: /root/hmr/
Desktop:   /home/dslab/hmr-lab/

map.yaml          map.pgm
map_fresh.yaml    map_fresh.pgm

Use: /root/hmr/map_fresh.yaml
```

Both map pairs were confirmed on the desktop. The fresh map was saved at about 16:38 on 7 November. Grid: 384 x 384 cells; resolution 0.05 m/cell; origin (-10, -10), identity orientation. The map has incomplete/doubled outlines. Localization remains unconfirmed.

Earlier WSL setup: Ubuntu 20.04 user shaha had Noetic and a successfully built hack_my_robot workspace. That does not establish that the same mission package exists in the lab Docker container. The old duplicate package-name build issue was resolved in WSL.

Current blocker: diagnose next

Observed: goals reach the desktop planner, but the Pi cannot contact the desktop move_base XML-RPC endpoint. Receiving scans or odometry on the desktop does not prove that the opposite connection direction works.

A. Desktop host terminal: inspect the listener

```
hostname -I
sudo ufw status
sudo docker inspect -f '{{.HostConfig.NetworkMode}}' hmr-noetic-gui
sudo ss -lntp | grep move_base
```

Already confirmed earlier: desktop IP 10.10.169.186, UFW inactive, Docker network host. The ss listener check is the next outstanding result. Record listening addresses and ports; ROS node ports change after restarts.

B. Pi SSH terminal: test the return connection

```
source /opt/ros/noetic/setup.bash
unset ROS_HOSTNAME
export ROS_IP=10.10.184.153
export ROS_MASTER_URI=http://10.10.184.153:11311

ping -c 3 10.10.169.186
rosnode info /move_base
timeout -s INT -k 2 8 rosnode ping -c 1 /move_base
```

The last ping-to-node attempt timed out. Capture the advertised URI from rosnode info and compare its host/port with the desktop listener. If necessary, test TCP to that exact port using Python socket.create_connection from the Pi. An ICMP ping result alone is not sufficient.

Interpretation and next decision

Wrong advertised IP: correct ROS_IP/ROS_HOSTNAME in the shell that launches the node, then restart that node. **Listener only on loopback:** investigate node environment/binding. **Correct listener but remote TCP blocked:** investigate host filtering, routing, and campus network policy with the lab administrator. Do not assume UFW is the only filter or disable protections blindly.

ROS1 uses the master plus node XML-RPC and dynamically assigned TCPROS ports. Making only port 11311 reachable is not enough. Running the navigation processes on the Pi is a possible fallback if resources/packages permit, but it was not attempted or validated.

Pass condition before autonomous testing

Pi rosnode ping /move_base succeeds; turtlebot3_core shows an incoming /cmd_vel connection to move_base; a bounded test command reaches the Pi; motors still respond. Then recheck localization and attempt a short clear navigation goal.

Restart recipe: Pi and motors

Use this after a shutdown only. If bringup is already running, keep it running. Verify current IPs first. Do not start duplicate copies of bringup or teleop.

1. Desktop terminal: connect to Pi

```
ssh ubuntu@10.10.184.153
```

2. Pi terminal: bringup

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
unset ROS_HOSTNAME
export ROS_IP=10.10.184.153
export ROS_MASTER_URI=http://10.10.184.153:11311
export TURTLEBOT3_MODEL=burger
export LDS_MODEL=LDS-02
roslaunch turtlebot3_bringup turtlebot3_robot.launch --screen
```

Leave running. Expected startup includes connecting to /dev/ttyACM0 at 115200 baud, topic setup, Connected to OpenCR board, and Calibration End. LiDAR was found on /dev/ttyUSB0 as LD08. Do not start a separate unrelated ROS master.

3. Separate Pi SSH terminal: manual test

First ensure autonomous navigation is stopped or its goals are cancelled, and no competing velocity publisher is active. Clear cables and floor space. The Pi was initially powered by USB alone; later the robot was switched to a battery. Battery charge and current motor supply must be checked.

```
source /opt/ros/noetic/setup.bash
unset ROS_HOSTNAME
export ROS_IP=10.10.184.153
export ROS_MASTER_URI=http://10.10.184.153:11311
export TURTLEBOT3_MODEL=burger
rostopic pub -1 /motor_power std_msgs/Bool "data: true"
roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch
```

Click inside teleop. Press w three times, wait about one second, then s. A brief a command can test turning, followed by s. Teleop does not provide automatic obstacle avoidance. Press s then Ctrl+C before returning to navigation.

What worked: movement resumed after motor_power=true. **What this does not prove:** that the desktop navigation publisher can reach the Pi. A teleop node launched over SSH actually runs on the Pi and avoids that cross-machine connection.

Restart recipe: desktop navigation

1. Desktop host: enter the existing container

```
sudo docker start hmr-noetic-gui
sudo docker exec -it hmr-noetic-gui bash
```

Do not run docker commands inside the container. If docker reports the name does not exist, recover the original container/image setup rather than creating a blank container and assuming it contains the prepared files.

2. Docker shell: start navigation

```
source /opt/ros/noetic/setup.bash
unset ROS_HOSTNAME
export ROS_IP=10.10.169.186
export ROS_MASTER_URI=http://10.10.184.153:11311
export TURTLEBOT3_MODEL=burger

rosparam set /move_base/local_costmap/always_send_full_costmap true
rosparam set /move_base/global_costmap/always_send_full_costmap true
roslaunch turtlebot3_navigation turtlebot3_navigation.launch \
  map_file:=/root/hmr/map_fresh.yaml open_rviz:=false
```

Leave running. RViz is deliberately started separately. Earlier, closing the RViz launched with navigation could shut down that launch and remove the map server.

3. If initialization/grid publication stalls again

Use a new sourced Docker shell, with the same ROS_IP and master settings. Leave Pi bringup, AMCL, map server, and robot_state_publisher running. The following selective restart worked in this session; it does not fix network reachability.

```
rosservice call /request_nomotion_update "{}"
rosnode kill /move_base
rosparam set /move_base/local_costmap/always_send_full_costmap true
rosparam set /move_base/global_costmap/always_send_full_costmap true
rosrun move_base move_base __name:=move_base
```

This rosrun invocation reuses parameters already loaded by the full navigation launch. It is not a replacement for loading navigation configuration on a fresh master. Do not run two move_base processes.

4. Verify from another sourced Docker shell

```
timeout -s INT -k 2 6 rostopic hz /move_base/local_costmap/costmap
timeout -s INT -k 2 6 rosrun tf tf_echo map base_footprint
```

RViz and a bounded goal test

Start RViz in another Docker shell

```
source /opt/ros/noetic/setup.bash
unset ROS_HOSTNAME
export ROS_IP=10.10.169.186
export ROS_MASTER_URI=http://10.10.184.153:11311
rviz -d /opt/ros/noetic/share/turtlebot3_navigation/rviz/turtlebot3_navigation.rviz \
  __name:=rviz_check
```

Display and pose checklist

Set Fixed Frame to map. Keep Map, RobotModel, and LaserScan visible. Collapse Local Map to find Amcl Particles; hide particles temporarily to distinguish them from LiDAR. The particles are possible poses, not obstacles. Reopen the sidebar using its edge triangle or Panels > Displays.

2D Pose Estimate: click and hold at the robot's actual physical map location, drag toward the direction its front faces, then release. Use one arrow only. Each new estimate replaces the old one. Arrow length is not travel distance. Do not assume the displayed robot is already in the correct location.

The operator identified the bin as a curved scan feature and the wall behind the robot as a straight scan feature. Last reported wall distance: about 1.11 m. The screen view was rotated several times, so earlier instructions such as drag left are not reusable. Compare the live scan against the saved black walls and recognizable physical landmarks.

```
rostopic info /initialpose
rosservice call /request_nomotion_update "{}"
```

Confirmed wiring: /rviz_check publishes /initialpose; /amcl subscribes. A successful service call or TF chain does not independently prove correct localization. Do not publish an invented static map-to-odom transform to hide a localization error.

Only after network and localization checks pass

Enable Local Map > Costmap. Confirm Map received, Map OK, Transform OK, and nonzero dimensions (previously 60 x 60 at 0.05 m/cell). Select 2D Nav Goal, choose a short clear destination ahead, drag for the final heading, and release. Observe physical travel and stopping. Very close goals may fall inside arrival tolerance; inspect status instead of repeatedly guessing.

Cancel and inspect

```
rostopic pub -1 /move_base/cancel actionlib_msgs/GoalID "{}"
timeout -s INT -k 2 5 rostopic echo -n 1 /move_base/status
timeout -s INT -k 2 5 rostopic echo /cmd_vel
```

Common status values: 0 pending, 1 active, 2 preempted, 3 succeeded, 4 aborted, 5 rejected. No movement plus a nonzero command requires delivery/torque checks. A zero command requires planner/status/log inspection. A cancelled goal is not evidence that the robot is physically stopped; verify stopping.

Troubleshooting evidence and cautions

Observation	Meaning / next check
IMU about 198 Hz, no sensor_state	Serial traffic exists. Do not conclude all OpenCR communications failed. Check motor/encoder communication and new bringup errors.
Publishers: None on cmd_vel	No registered command publisher at that instant. Teleop must remain running while inspecting.
Teleop prints speed	Only proves key handling. Observe cmd_vel on Pi and check core connections.
Costmap footprint 10 Hz, grid absent	Update loop was alive. We enabled full grids and selectively restarted move_base.
Legacy Pre-Hydro warnings	Initialization proceeded despite them. Later local layer configuration was reported through legacy parameters; inspect loaded configuration if needed.
Master accessible, move_base unreachable	ROS graph access is not equivalent to direct node access. Check advertised address and dynamic node ports.
Green arrows surrounding robot	AMCL particle cloud. Hide it for scan comparison; do not treat it as laser returns.
No map received in disabled display	Disabled displays may retain stale status. Re-enable and inspect dimensions and actual incoming messages.

Useful Pi diagnostics

```
ls -l /dev/ttyACM* /dev/ttyUSB*
timeout -s INT -k 2 5 rostopic hz /imu
timeout -s INT -k 2 5 rostopic echo -n 1 /sensor_state
timeout -s INT -k 2 5 rostopic echo -n 1 /battery_state
rosnode info /turtlebot3_core
```

The firmware source inspected during the session skips publishing sensor_state when wheel encoder reading fails. That makes wheel communication a plausible secondary issue, not a confirmed diagnosis for this specific robot. Do not repeatedly reflash firmware without evidence.

Do not carry these assumptions forward

Do not infer clock skew from two date commands separated by a password prompt. Do not assume battery percentage is calibrated. Do not assume all navigation failures were caused by power. Do not install ROS1 Focal packages natively into the Noble host. Do not apply commands copied for WSL to the lab desktop shell.

Complete the route-change criterion

The following is the continuation plan. It has not been executed end to end. Navigation recovery alone does not demonstrate the competition route-change criterion.

1. Confirm the mission package on the lab machine

```
rospack find hack_my_robot
rospack find turtlebot3_navigation
rospack find turtlebot3_slam
```

If hack_my_robot is missing, find the intended lab workspace or obtain the competition repository, install its dependencies, build the workspace, and source its devel/setup.bash. WSL previously built it in ~/catkin_ws/src/hack_my_robot_2026; installation in this container is unconfirmed. Avoid duplicate package names.

2. Inspect launch and node before sending mission messages

Repository supplied by the operator: https://github.com/SMART-NYUAD/hack_my_robot_2026 . Its live contents were not retrievable while preparing this handoff. Use the local checkout as the authority for current topic types and behavior.

The supplied hack_my_robot_complete.launch includes turtlebot3_navigation and launches hack_my_robot_node. It hardcodes map_file=/home/sxp8070/map.yaml. Change the launch configuration or expose a map argument to use /root/hmr/map_fresh.yaml in this container. Do not launch a second navigation stack alongside the one already running.

```
roscd hack_my_robot
cat launch/hack_my_robot_complete.launch
rostopic info /addpose
rostopic info /waypoints
rostopic info /path_ready
rostopic info /start_journey
```

Reported mission procedure: choose three waypoints through a pose tool remapped to /addpose; /path_ready runs once and /start_journey loops. Verify this in the actual node and README. Do not guess message types: /addpose may be PoseWithCovarianceStamped, and the trigger topics were reported as std_msgs/Empty. During localization the normal pose tool must still publish /initialpose to AMCL.

3. Demonstrate a controlled route change

First establish a reproducible baseline route and record the waypoint coordinates. Then submit a different valid route through the verified mission interface, keep normal obstacle avoidance enabled, and record the physical robot following the changed route without collision. Confirm the evaluator's exact requirements and scoring from the current official rubric.

4. Evidence to hand over

Retain the map pair, launch changes, mission code/version, graph/topic types, baseline and changed waypoints, logs, a short video showing the changed path, and the relevant AI interaction history required by the event. This summary is not a substitute for a complete prompt/answer history if the organizers require it.

Transfer checklist and operator brief

Transfer these existing assets from the lab desktop

The actual map files and container image are on the lab desktop, not embedded in this PDF. Verify that they still exist before deleting or recreating containers. The commands below create a portable backup; they were not run during preparation of this document.

```
# Desktop host: verify maps and container mounts
ls -lh ~/hmr-lab/map*.yaml ~/hmr-lab/map*.pgm
sudo docker inspect -f '{{json .Mounts}}' hmr-noetic-gui

# Optional map backup for the next operator
tar -czf ~/criterion3-maps.tar.gz -C ~/hmr-lab \
  map.yaml map.pgm map_fresh.yaml map_fresh.pgm

# Optional image snapshot: can be large; excludes bind-mounted files
sudo docker commit hmr-noetic-gui hmr-noetic:handoff
sudo docker save -o ~/hmr-noetic-handoff.tar hmr-noetic:handoff
```

Transfer the map archive separately from the image archive. Also retain container startup settings, X11 display permissions/mounts, any mission workspace, and local changes. No passwords are included in this PDF; obtain credentials directly from the operator. On a new desktop the host IP, GUI setup, and ROS_IP must be updated.

Brief for the next person or assistant

Authorized TurtleBot3 Criterion 3.1 practice. Pi Ubuntu20.04/Noetic Burger IP 10.10.184.153. Lab desktop Ubuntu24.04 IP 10.10.169.186 uses Docker hmr-noetic-gui, host network, UFW inactive. Map /root/hmr/map_fresh.yaml is mounted from /home/dslab/hmr-lab. OpenCR v1.2.6 connects; IMU about 198Hz. sensor_state repeatedly absent. Teleop worked after motor_power=true. Costmap publication recovered by setting always_send_full_costmap on both costmaps and restarting only move_base after AMCL was running. Localization is not confirmed. RViz goal reaches move_base and it emits angular.z=1.0, but robot stays stationary. Pi cannot rosnode ping move_base; core has no incoming cmd_vel link. Goal cancelled. Diagnose Pi-to-desktop node connectivity first. Outstanding outputs: desktop ss listener, Pi ping desktop and rosnode info move_base. Do not restart mapping or assume changing pose alone fixes this. Mission package availability in Docker is unconfirmed. No scored Criterion3 route change completed.

Reference material

Primary references used during the session, plus the operator's terminal logs/screenshots:
ROBOTIS TurtleBot3 OpenCR documentation:
https://emanual.robotis.com/docs/en/platform/turtlebot3/appendix_opencr1_0/
ROBOTIS OpenCR firmware repository:
https://github.com/ROBOTIS-GIT/OpenCR
Competition repository supplied by operator:
https://github.com/SMART-NYUAD/hack_my_robot_2026

The original lab logs and screenshots remain the evidence for observed results. Commands labelled as future steps or optional backups have not been validated in a fresh environment. Network filtering, current battery state, and successful autonomous navigation remain open questions.